LECTURE FOUR

ASSIGNING COSTS TO PRODUCTS

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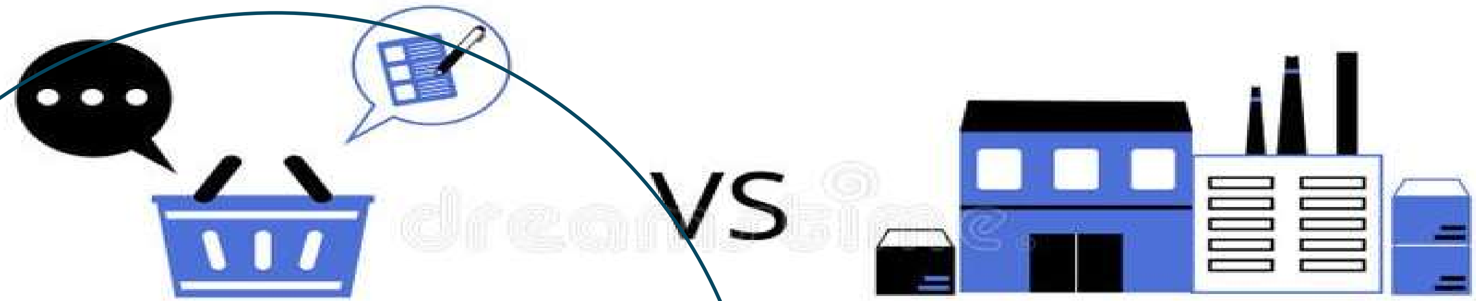
Two Cost Accounting Systems



Job Order Costing

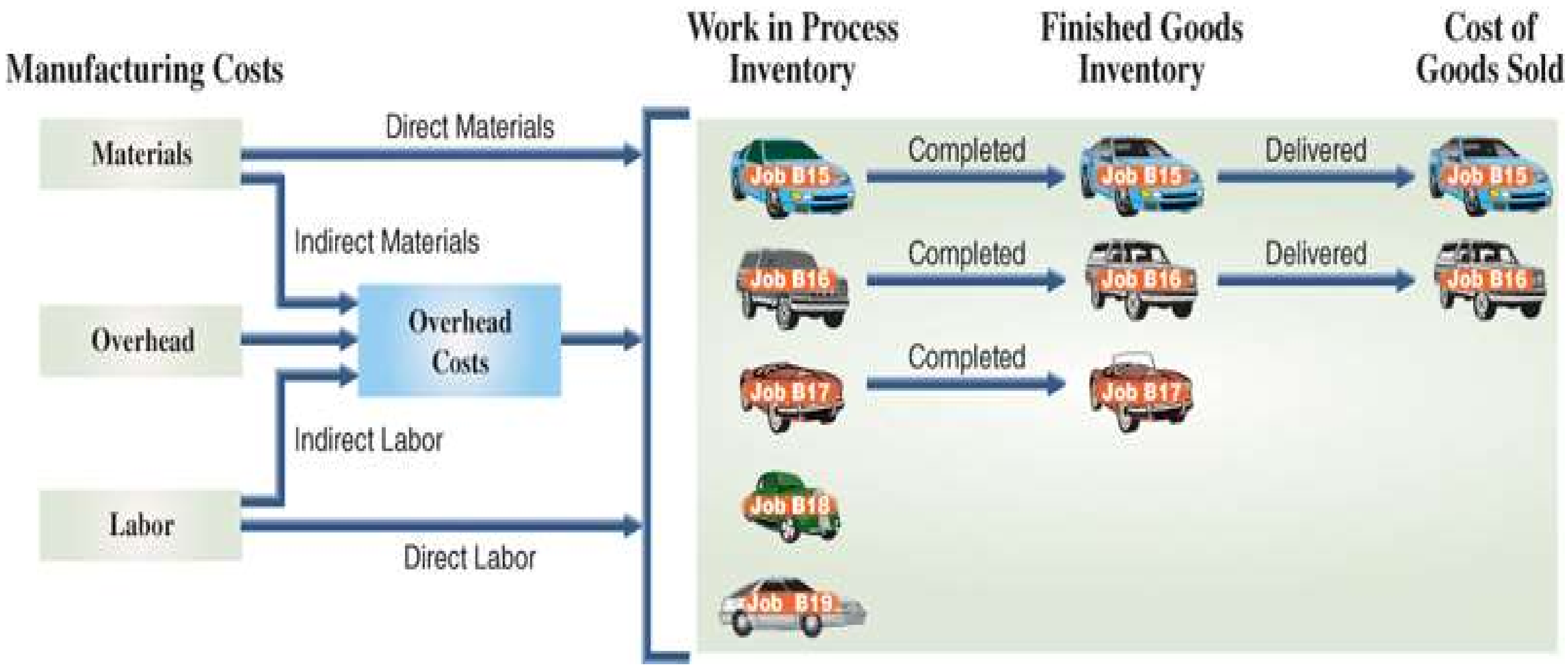
- Custom, unique jobs
- Production by order
- Low Volume
- Low Standardization
- E.g. Customized furniture & Car manufacturing companies

Process Costing

- Continuous, identical units
- Mass production
- High Volume
- High Standardization
- E.g.: Paper manufacturing company
Water bottle manufacturing

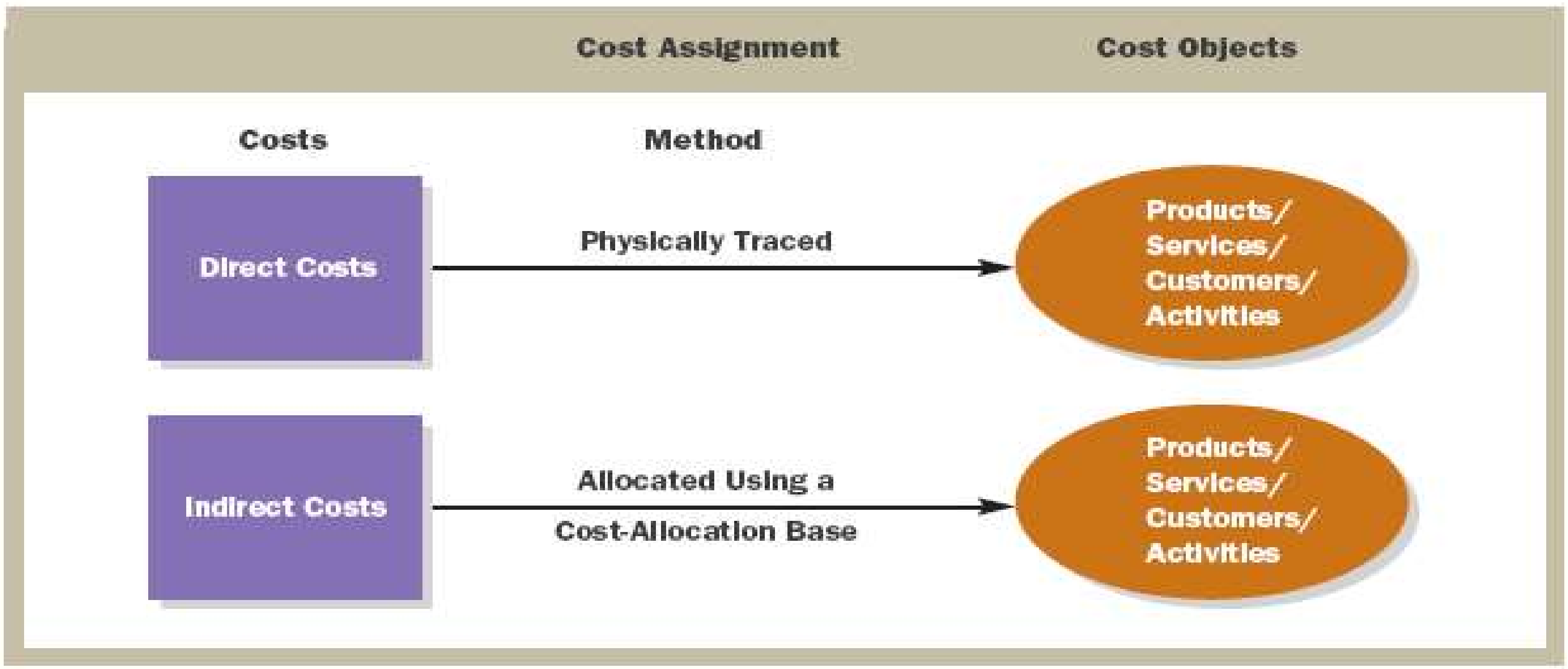
Job-Order Costing (Production Cost Flow)

Example: Car Manufacturing Company:



Cost Assignment

Direct costs are physically traced to a cost object. Indirect costs are allocated using a cost-allocation base.



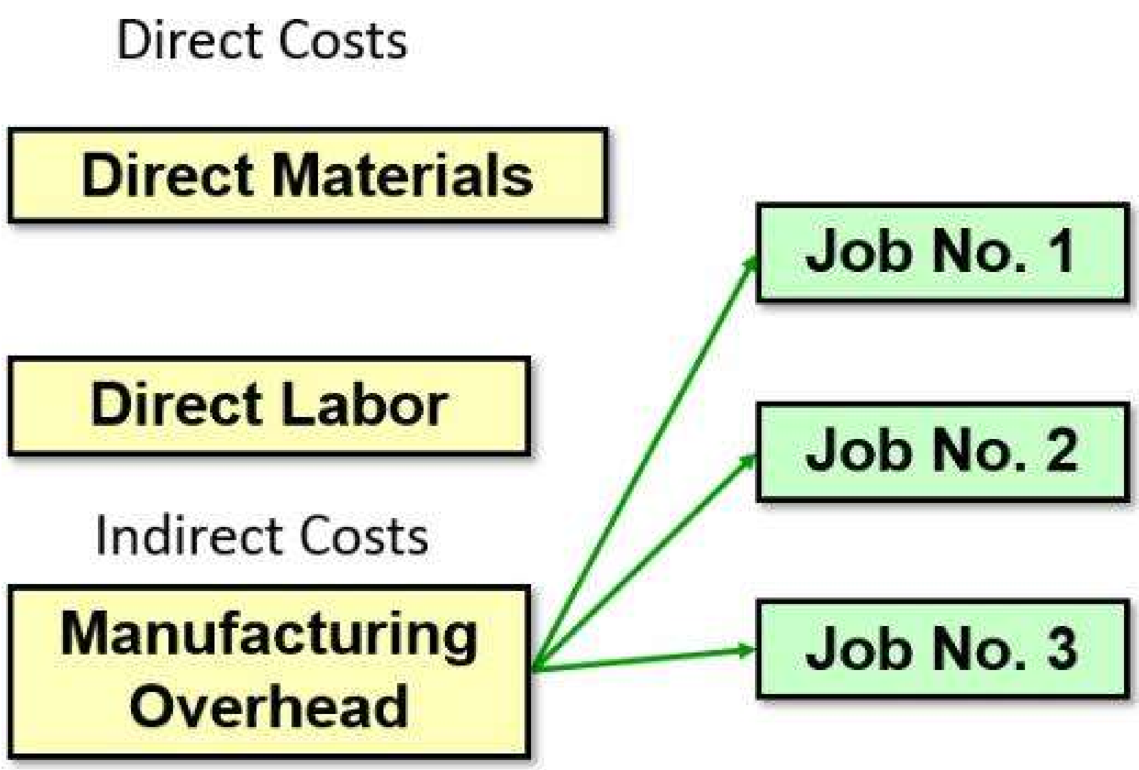
<u>Components of Manufacturing Cost</u>

1) **Direct Materials (DM):** raw inputs used in production (Traceable for each job order)

2) **Direct Labor (DL):** wages of production workers
(Traceable for each job order)

3) **Manufacturing Overhead (MOH):** indirect costs (rent, depreciation, etc.) (Not Traceable for each job order !!) because it is **shared** across all jobs.

Job-Order Costing – Cost Flow 2



Manufacturing Overhead, including *indirect materials* and *indirect labor,* are allocated to all jobs rather than directly traced to each job.

The Job Cost Sheet

PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-17

Date Completed

Department B3

Units Completed

Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount

Cost Summary	Units Shipped		
	Date	Number	Balance
Direct Materials			
Direct Labor			
Manufacturing Overhead			
Total Cost			
Unit Product Cost			

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Measuring Direct Materials Cost – Part 1

PearCo Materials Requisition Form

Requisition No. X7 - 6890

Date 3-4-17

Job No. A - 143

Department B3

Description	Quantity	Unit Cost	Total Cost
2 x 4, 12 feet	12	$ 3.00	$ 36.00
1 x 6, 12 feet	20	4.00	80.00
			$ 116.00

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Measuring Direct Materials Cost – Part 2

PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-17

Date Completed

Department B3

Units Completed

Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116						

Cost Summary		Units Shipped		
Direct Materials	$ 116	Date	Number	Balance
Direct Labor				
Manufacturing Overhead				
Total Cost				
Unit Product Cost				

Measuring Direct Labor Costs

PearCo Employee Time Ticket

Time Ticket No. 36 | Date 3-5-17

Employee I. M. Skilled | Station 42

Starting Time	Ending Time	Hours Completed	Hourly Rate	Amount	Job No.
0800	1600	8.00	$ 15.00	$ 120.00	A-143
Totals		8.00	$ 15.00	$ 120.00	A-143

Supervisor C. M. Workman

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Job-Order Cost Accounting

PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-17

Date Completed

Units Completed

Department B3

Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120			

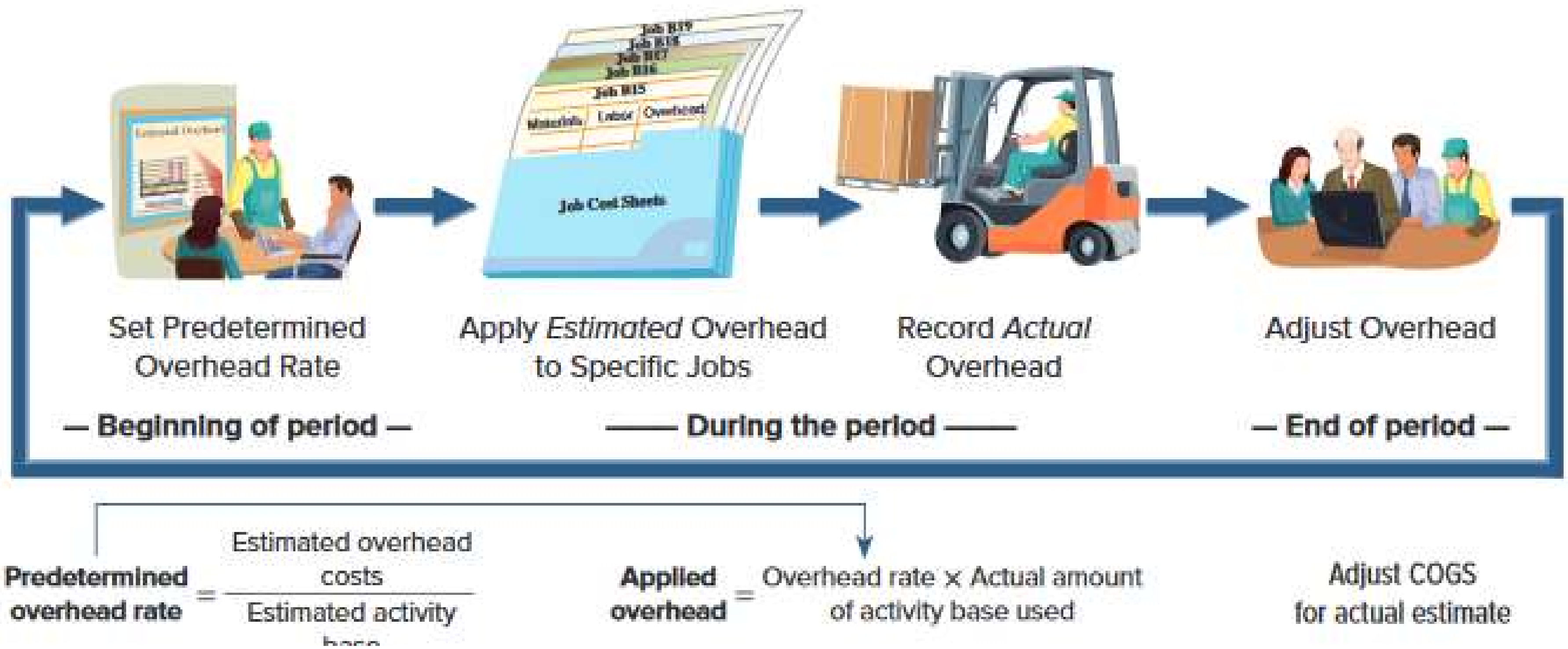
Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead				
Total Cost				
Unit Product Cost				

Learning Objective 1

Compute a predetermined overhead rate.

Overhead Costs

Exhibit 19.11



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Why Use an Allocation Base?

An allocation base, such as direct labor hours, direct labor dollars, or machine hours, is used to assign manufacturing overhead to individual jobs.

We use an allocation base because:

a. It is impossible or difficult to trace overhead costs to particular jobs.

b. Manufacturing overhead consists of many different items ranging from the grease used in machines to the production manager's salary.

c. Many types of manufacturing overhead costs are fixed even though output fluctuates during the period.

Application of Factory Overhead)

The predetermined factory overhead (OH) rate used to apply overhead to jobs is determined *before* the period begins.

$$\text{OH Rate} = \frac{\text{Estimated factory overhead amount for the year}}{\text{Estimated level of cost driver for the year}}$$

Some Possible Cost Drivers:
1. direct labor-hours
2. machine-hours
3. number of set-ups
4. number of orders
5. manufacturing cycle-time

The OH rate can be calculated on a plantwide or a departmental basis.

Application of Factory Overhead (continued)

$$\text{OH Rate} = \frac{\text{Predetermined factory overhead amount for the year}}{\text{Predetermined level of cost driver for the year}}$$

Overhead applied = OH Rate × Actual activity

Based on estimates, and determined *before* the period begins

Actual amount of the allocation base, such as direct labor-hours, incurred during the period

The Need for a POHR

Predetermined overhead rates that rely upon estimated data are often used because:

1. Actual overhead for the period is not known until the end of the period, thus inhibiting the ability to estimate job costs during the period.
2. Actual overhead costs can fluctuate seasonally, thus misleading decision makers.

Learning Objective 2

Apply overhead cost to jobs using a predetermined overhead rate.

Overhead Application Rate

$$\text{POHR} = \frac{\$640{,}000 \text{ estimated total manufacturing overhead}}{160{,}000 \text{ estimated direct labor hours (DLH)}}$$

POHR = $4.00 per direct labor-hour

Recording Manufacturing Overhead

PearCo Job Cost Sheet

Job Number A - 143

Department B3

Item Wooden cargo crate

Date Initiated 3-4-17

Date Completed 3-5-17

Units Completed 2

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary	
Direct Materials	$ 116
Direct Labor	$ 120
Manufacturing Overhead	$ 32
Total Cost	
Unit Product Cost	

Units Shipped		
Date	Number	Balance

Calculating Total Cost of Job

PearCo Job Cost Sheet

Job Number A - 143
Date Initiated 3-4-17
Date Completed 3-5-17
Department B3
Units Completed 2
Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead	$ 32			
Total Cost	$ 268			
Unit Product Cost				

Calculating Unit Product Cost

PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-17

Date Completed 3-5-17

Units Completed 2

Department B3

Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead	$ 32			
Total Cost	$ 268			
Unit Product Cost	$ 134			

Quick Check 1

Job WR53 at NW Fab, Inc. required $200 of direct materials and 10 direct labor hours at $15 per hour. Estimated total overhead for the year was $760,000 and estimated direct labor hours were 20,000. What would be recorded as the cost of job WR53?

a. $200.

b. $350.

c. $380.

d. $730.

Quick Check 1a

Job WR53 at NW Fab, Inc. required $200 of direct materials and 10 direct labor hours at $15 per hour. Estimated total overhead for the year was $760,000 and estimated direct labor hours were 20,000. What would be recorded as the cost of job WR53?

a. $200.

b. $350.

c. $380.

d. Answer: $730.

POHR = $760,000/20,000 hours		$38
Direct materials		$200
Direct labor	$15 x 10 hours	$150
Manufacturing overhead	$38 x 10 hours	$380
Total cost		$730

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Job-Order Costing – A Managerial Perspective – Part 1

Inaccurately assigning manufacturing costs to jobs adversely influences planning and decisions made by managers.

1. Job-order costing systems can accurately trace ***direct*** materials and ***direct*** labor costs to jobs.
2. Job-order costing systems often fail to accurately allocate the manufacturing overhead costs used during the production process to their respective jobs.

Job-Order Costing – A Managerial Perspective – Part 2

Choosing an Allocation Base

Job-order costing systems often use allocation bases that do not reflect how jobs actually use overhead resources. The allocation base in the predetermined overhead rate must **drive** the overhead cost to improve job cost accuracy. A **cost driver** is a factor that causes overhead costs.

Many companies use a single predetermined **plantwide overhead rate** to allocate all manufacturing overhead costs to jobs based on their usage of direct-labor hours.

1. It is often **overly simplistic** and incorrect to assume that direct-labor hours is a company's *only* manufacturing overhead cost driver.
2. If more than one overhead cost driver can be identified, job cost accuracy is improved by using **multiple predetermined overhead rates**.

Job-Order Costing for Financial Statements to External Parties

The amount of overhead applied to all jobs during a period will differ from the actual amount of overhead costs incurred during the period.

1. When a company applies less overhead to production than it actually incurs, it creates what is known as underapplied overhead.
2. When it applies more overhead to production than it actually incurs, it results in overapplied overhead.

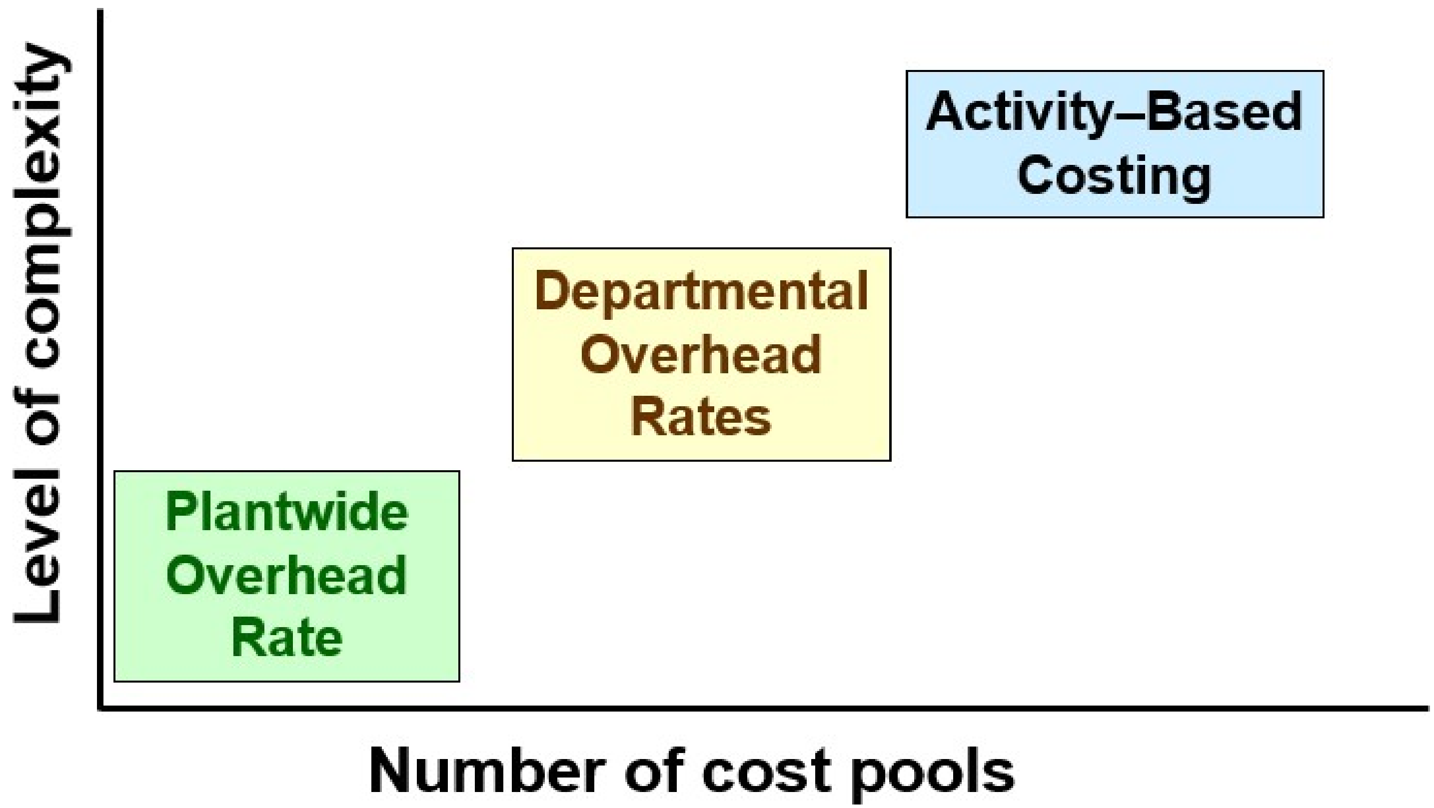
Financial Adjust for Overhead Applied

The cost of goods sold reported on a company's income statement must be adjusted to reflect underapplied or overapplied overhead.

1. The adjustment for underapplied overhead **increases cost of goods sold** and decreases net operating income.
2. The adjustment for overapplied overhead **decreases cost of goods sold** and increases net operating income.

Types of Overhead application rates

ABC differs from traditional cost accounting because numerous overhead cost pools are used.



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Multiple Predetermined Overhead Rates—An Activity-Based Approach

When a company creates overhead rates based on the activities that it performs, it is employing an approach called *activity-based costing.*

Activity-based costing is an alternative approach to developing multiple predetermined overhead rates. Managers use activity-based costing systems to more accurately measure the demands that jobs, products, customers, and other cost objects make on overhead resources.

Key Definitions and Concepts

The Five Steps for Implementing ABC

Baxter Battery Company
Income Statement
Year Ended December 31, 2017

Sales		$ 50,000,000
Cost of goods sold		
Direct materials	$ 15,000,000	
Direct labor	12,000,000	
Manufacturing overhead	14,000,000	41,000,000
Gross margin		9,000,000
Selling and administrative expenses		
Shipping expense	3,000,000	
Marketing expense	2,000,000	
General administrative expense	6,000,000	11,000,000
Net operating loss		$ (2,000,000)

The company makes two types of automobile batteries—SureStart (a standard battery) and LongLife (a deluxe battery). Baxter reported its first loss ever.

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Define Activities, Activity Cost Pools, and Activity Measures

At Baxter Battery, the ABC team selected the following activity cost pools and activity measures:

Activity Cost Pools at Baxter Battery		
Activity Cost Pool		Activity Measure
Customer orders	→	Number of customer orders
Design changes	→	Number of design changes
Order size	→	Machine-hours
Customer relations	→	Number of active customers
Other	→	Not applicable

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Define Activities, Activity Cost Pools, and Activity Measures – Part 2

Customer Orders - assigned all costs of resources that are consumed by taking and processing customer orders.

Design Changes - assigned all costs of resources consumed by customer requested design changes.

Order Size - assigned all costs of resources consumed as a consequence of the number of units produced.

Customer Relations - assigned all costs associated with maintaining relations with customers.

Other - assigned all organization-sustaining costs and unused capacity costs.

Assign costs to cost pools using a first-stage allocation.

Assign Overhead Costs to Activity Cost Pools –

Direct materials, direct labor, and shipping are excluded because Baxter Battery's existing cost system can directly trace these costs to products or customer orders.

Overhead Costs at Baxter Battery (Manufacturing and Nonmanufacturing)

Production Department		
Indirect factory wages	$ 6,000,000	
Factory equipment depreciation	3,500,000	
Factory utilities	2,500,000	
Factory building lease	2,000,000	$ 14,000,000
General Administrative Department		
Administrative wages and salaries	4,000,000	
Office equipment depreciation	900,000	
Administrative building lease	1,100,000	6,000,000
Marketing Department		
Marketing wages and salaries	1,500,000	
Selling expenses	500,000	2,000,000
Total overhead costs		$ 22,000,000

Assign Overhead Costs to Activity Cost Pools

	Activity Cost Pools					
	Customer Orders	Design changes	Order Size	Customer Relations	Other	Total
Production Department						
Indirect factory wages	$ 1,800,000	$ 1,800,000	$ 1,200,000	$ 600,000	$ 600,000	$ 6,000,000
Factory equipment depreciation	700,000	350,000	2,100,000	-	350,000	3,500,000
Factory utilities	-	250,000	1,500,000	-	750,000	2,500,000
Factory building lease	-	-	-	-	2,000,000	2,000,000
General Administrative Department						
Administrative wages and salaries	1,200,000	400,000	400,000	1,200,000	800,000	4,000,000
Office equipment depreciation	270,000	90,000	-	180,000	360,000	900,000
Administrative building lease	-	-	-	-	1,100,000	1,100,000
Marketing Department						
Marketing wages and salaries	450,000	150,000	-	750,000	150,000	1,500,000
Selling expenses	100,000	-	-	350,000	50,000	500,000
Total	$ 4,520,000	$ 3,040,000	$ 5,200,000	$ 3,080,000	$ 6,160,000	$ 22,000,000

Compute activity rates for cost pools.

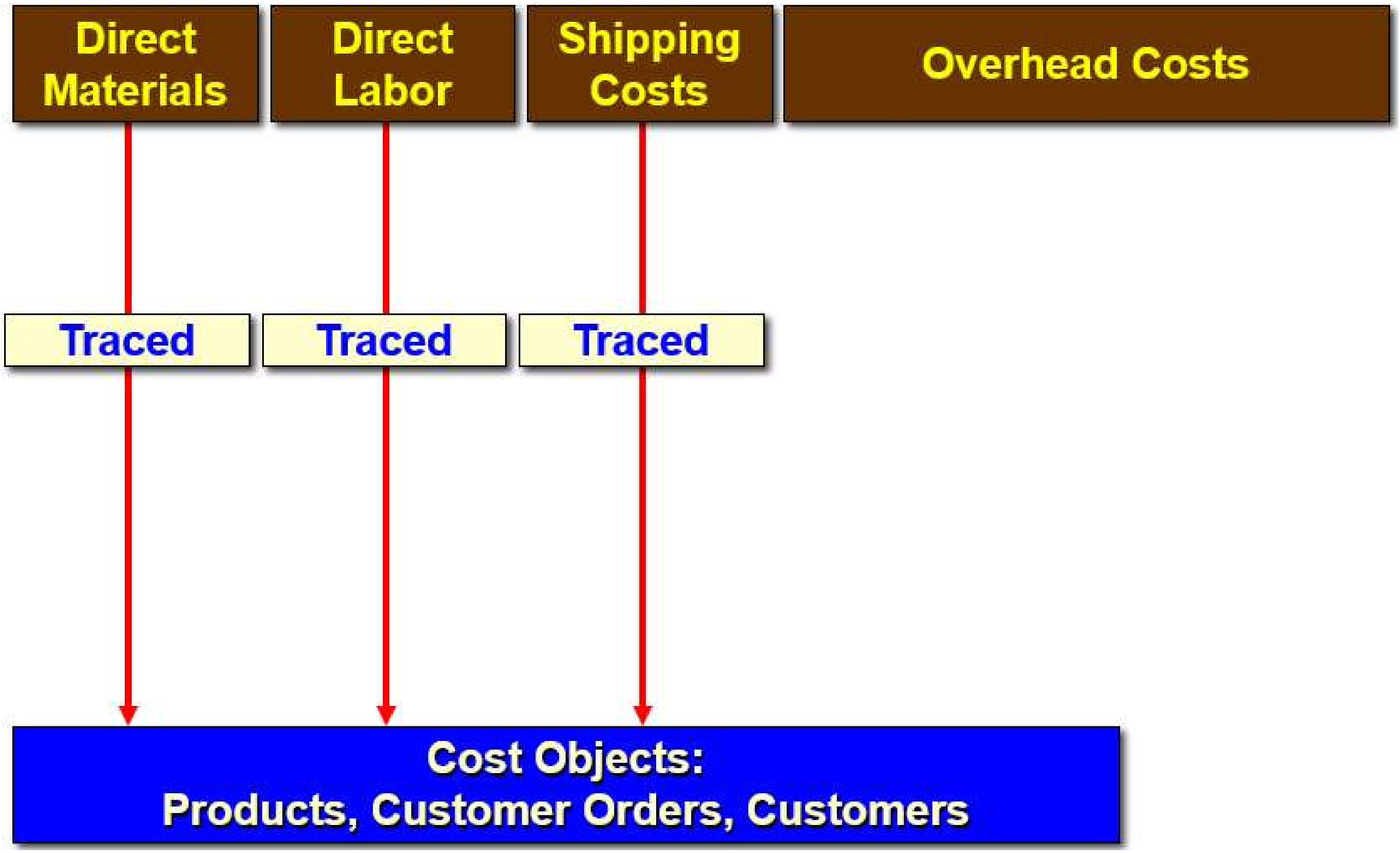
Calculate Activity Rates

The ABC team determines that Baxter Battery will have these total activities for each activity cost pool:

- 10,000 customer orders,
- 4,000 design changes,
- 800,000 machine-hours,
- 2,000 customers served.

Now the team can compute the individual activity rates by dividing the total cost for each activity by the total activity levels.

Calculate Activity Rates

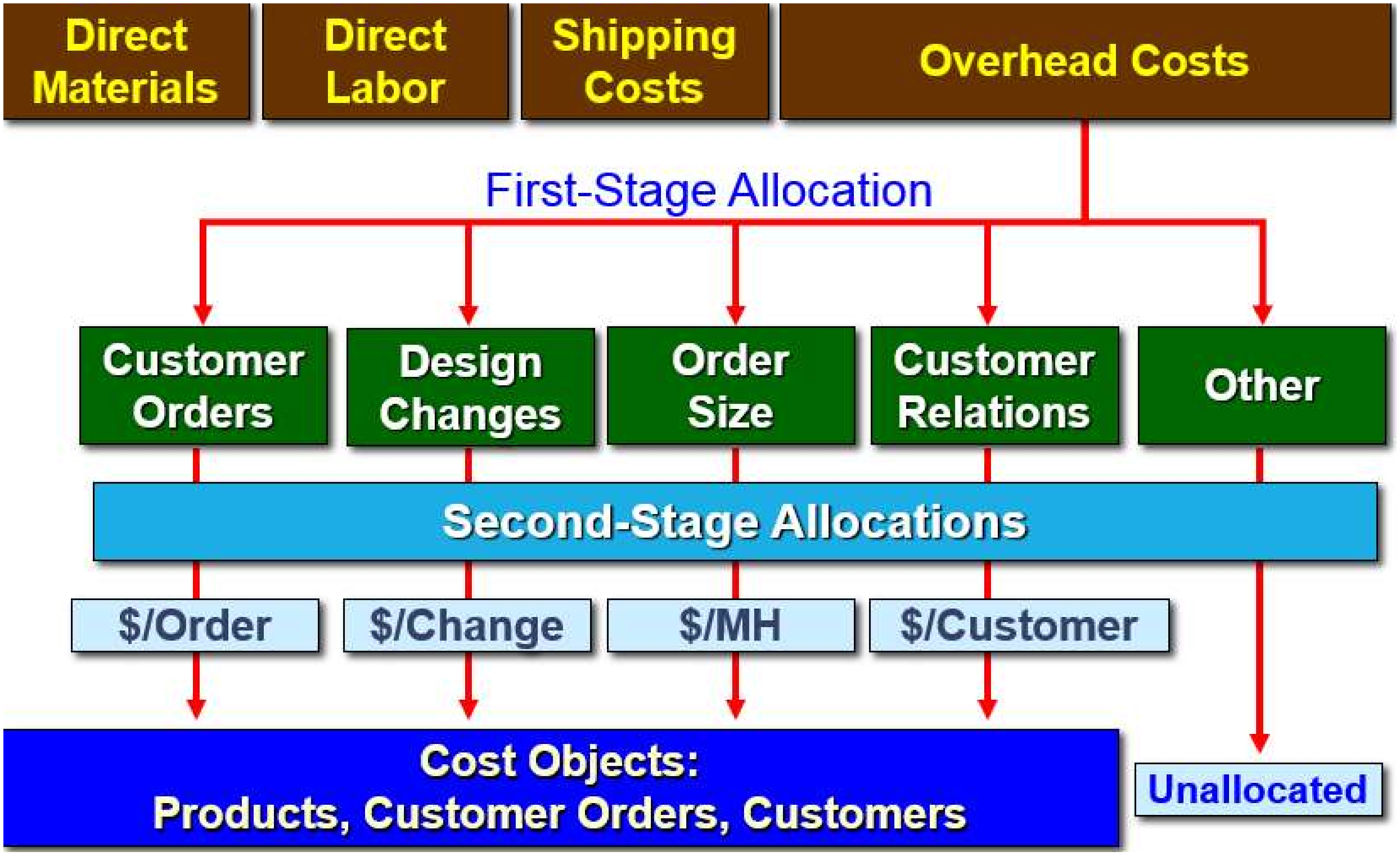


Calculate Activity Rates

Cost Objects:

Products, Customer Orders, Customers

Calculate Activity Rates



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Calculate Activity Rates

Computation of Activity Rates			
	(a)	(b)	(a) ÷ (b)
Activity Cost Pools	*Total Cost*	*Total Activity*	*Activity Rate*
Customer orders	$ 4,520,000	10,000 orders	$452 per order
Design changes	3,040,000	4,000 changes	$760 per change
Order size	5,200,000	800,000 MHs	$6.50 per MH
Customer relations	3,080,000	2,000 customers	$1,540 per customer
Other	6,160,000	Not applicable	Not applicable
Total	$ 22,000,000		

These are organization-sustaining costs and will not be assigned to products or customers.

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Assigning Overhead to Products

Baxter Battery Information

SureStart

1. Requires no new design resources.
2. 800,000 batteries ordered with 4,000 separate orders.
3. Each SureStart requires 36 minutes of machine time for a total of 480,000 machine-hours.

LongLife

1. Requires new design resources.
2. 400,000 batteries ordered with 6,000 separate orders.
3. 4,000 custom designs prepared.
4. Each LongLife requires 48 minutes of machine time for a total of 320,000 machine-hours.

Assigning Overhead to Products

Overhead Cost for the SureStart			
	(a)	(b)	(a) × (b)
Activity Cost Pools	*Activity Rate*	*Activity*	*ABC Cost*
Customer orders	$ 452.00	4,000	$ 1,808,000
Design changes	760.00	-	-
Order size	6.50	480,000	3,120,000
Total			$ 4,928,000

Overhead Cost for the LongLife			
	(a)	(b)	(a) × (b)
Activity Cost Pools	*Activity Rate*	*Activity*	*ABC Cost*
Customer orders	$ 452.00	6,000	$ 2,712,000
Design changes	760.00	4,000	3,040,000
Order size	6.50	320,000	2,080,000
Total			$ 7,832,000

$4,928,000 + $7,832,000 + $9,240,000 (not assigned) = $22,000,000

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Use activity-based costing to compute product and customer margins.

Prepare Management Reports

Product Margin Calculations

The first step in computing product margins is to gather each product’s sales and direct cost data.

	SureStarts	LongLifes	Total
Sales	$ 31,300,000	$ 18,700,000	$ 50,000,000
Direct costs			
Direct material	9,000,000	6,000,000	15,000,000
Direct labor	7,000,000	5,000,000	12,000,000
Shipping	2,000,000	1,000,000	3,000,000

Prepare Management Reports

Product Margin Calculations

The second step in computing product margins is to incorporate the previously computed **activity-based cost assignments** pertaining to each product.

	SureStarts	LongLifes	Total
Sales	$ 31,300,000	$ 18,700,000	$ 50,000,000
Direct costs			
Direct material	9,000,000	6,000,000	15,000,000
Direct labor	7,000,000	5,000,000	12,000,000
Shipping	2,000,000	1,000,000	3,000,000
ABC cost assignments			
Customer orders	1,808,000	2,712,000	4,520,000
Design changes		3,040,000	3,040,000
Order size	3,120,000	2,080,000	5,200,000

Prepare Management Reports

Product Margin Calculations

The third step in computing product margins is to deduct each product's direct and indirect costs from sales.

	SureStarts		LongLifes	
Sales		$ 31,300,000		$ 18,700,000
Costs				
Direct material	$ 9,000,000		$ 6,000,000	
Direct labor	7,000,000		5,000,000	
Shipping	2,000,000		1,000,000	
Customer orders	1,808,000		2,712,000	
Design changes			3,040,000	
Order size	3,120,000		2,080,000	
Total cost		22,928,000		19,832,000
Product margin		$ 8,372,000		$ (1,132,000)

Prepare Management Reports

Product Margin Calculations

The product margins can be reconciled with the company's net operating income as follows:

	SureStarts	LongLifes	Total
Sales	$ 31,300,000	$ 18,700,000	$ 50,000,000
Total costs	22,928,000	19,832,000	42,760,000
Product margins	$ 8,372,000	$ (1,132,000)	$ 7,240,000
Less costs not assigned to products:			
Customer relations			3,080,000
Other			6,160,000
Total			9,240,000
Net operating loss			$ (2,000,000)

Product Margins Computed Using the Traditional Cost System

The second step in computing product margins is to compute the plantwide overhead rate.

Manufacturing Overhead Costs at Baxter Battery

Production Department	
Indirect factory wages	$ 6,000,000
Factory equipment depreciation	3,500,000
Factory utilities	2,500,000
Factory building lease	2,000,000
Total manufacturing overhead	$ 14,000,000

$$\text{Plantwide manufacturing overhead rate} = \frac{\$14{,}000{,}000}{800{,}000 \text{ MH}} = \$17.50 \text{ per machine-hour}$$

	Machine-hours
SureStarts (800,000 @ 0.60 hours)	480,000
LongLifes (400,000 @ 0.80 hours)	320,000
Total machine-hours	800,000

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Product Margins Computed Using the Traditional Cost System

The third step in computing product margins is to allocate manufacturing overhead to each product.

	Machine Hours	Overhead Rate	Overhead Allocated
SureStarts	480,000	$ 17.50	$ 8,400,000
LongLifes	320,000	17.50	5,600,000
Total overhead allocated to products			$ 14,000,000

480,000 hours × $17.50 per hour = $8,400,000

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Product Margins Computed Using the Traditional Cost System

	SureStarts		LongLifes		Total	
Sales		$ 31,300,000		$ 18,700,000		$ 50,000,000
Cost of goods sold						
Direct materials	$ 9,000,000		$ 6,000,000		$ 15,000,000	
Direct labor	7,000,000		5,000,000		12,000,000	
Manufacturing overhead	8,400,000	24,400,000	5,600,000	16,600,000	14,000,000	41,000,000
Product margin		$ 6,900,000		2,100,000		9,000,000
Selling and administrative						11,000,000
Net operating loss						$ (2,000,000)

Shipping expense	$ 3,000,000
Marketing expense	2,000,000
General administrative expense	6,000,000
	$ 11,000,000

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Differences Between ABC and Traditional Product Costs

	SureStarts	LongLifes
Product margins - traditional	$ 6,900,000	$ 2,100,000
Product margins - ABC	8,372,000	(1,132,000)
Change in reported margins	$ 1,472,000	$ (3,232,000)

The traditional cost system overcosts the Sure Starts and reports a lower product margin for this product.

The traditional cost system undercosts the LongLifes and reports a higher product margin for this product.

Differences Between ABC and Traditional Product Costs

There are three reasons why the reported product margins for the two costing systems differ from one another.

1-Traditional costing allocates all manufacturing overhead to products. ABC costing only assigns manufacturing overhead costs consumed by products to those products.

Differences Between ABC and Traditional Product Costs

2-Traditional costing allocates all manufacturing overhead costs using a volume-related allocation base. ABC costing also uses non-volume related allocation bases.

3-Traditional costing disregards selling and administrative expenses because they are assumed to be period expenses. ABC costing directly traces shipping costs to products and includes nonmanufacturing overhead costs caused by products in the activity cost pools that are assigned to products.